

Part 3: Product, API & Lifecycle Thinking

Question 1

In your own words, explain how moving from a synchronous API request (waiting for an immediate response) to an asynchronous webhook model changes both the technical data flow and the user experience during onboarding.

The Problem with the Current (Sync) Flow

The app sends a request to FinVerify and freezes — waiting up to 15+ seconds. If the API times out, the user sees a blank loading screen and drops off. One open connection blocks everything.

How the Async Webhook Model Works

The frontend sends the documents and immediately gets back a simple "received" acknowledgment. The connection closes. FinVerify processes in the background and, once done, fires a webhook — a callback to our server — with the result (pass/fail). Our backend then pushes a notification to the user.

Technical Changes

- No idle connection: Server is freed instantly; no timeout risk.
- State tracking: Backend stores a 'KYC pending' status per user, updated when the webhook fires.
- Event-driven: Result flows as FinVerify → Webhook → Our server → Push/SMS to user.

User Experience Change

Instead of a frozen screen, users see: "We are verifying your documents — you can close this screen and we'll notify you." They leave the app freely and return only when the result is ready. Drop-offs from timeouts are eliminated.

Question 2

If we transition to this asynchronous model, users will leave the app while their KYC is processing. Name three new product metrics we will need to track to measure onboarding health under this new lifecycle flow.

With users exiting mid-funnel, the old session-based funnel metrics no longer apply. These three new metrics cover the gaps:

1. Webhook-to-Activation Rate

What it measures: % of users who return to the app and complete activation after receiving their KYC result.

Why it matters: This is the new conversion metric for the async funnel. A drop here means notifications aren't compelling enough or re-entry is confusing. **Target:** >75%.

2. Notification Click-Through Rate (CTR)

What it measures: % of result notifications (push/SMS) that are delivered and tapped to re-open the app.

Why it matters: If users never see or tap the notification, the async model fails silently. Low delivery = permission issues. Low CTR = weak copy or wrong channel.

3. KYC Pending Abandonment Rate

What it measures: % of users whose result was delivered but who never returned within 72 hours.

Why it matters: This replaces 'loading screen abandonment' as the primary drop-off signal and triggers a secondary re-engagement campaign.

Question 3

For users who fail KYC due to a blurry image upload (failed_clear_error), outline a data-driven automated engagement strategy (timing of push notifications/SMS, in-app copy hints) to bring them back into the app to retry. How would you measure the success of this intervention?

These users are not fraudulent — they just uploaded a bad photo. The fix is simple: reach them fast, guide them clearly, and remove every possible friction from retrying. The plan runs over three weeks.

Week 1 — Immediate Re-engagement (Days 1–7)

Day 1 (within 1 hr): Push notification: "Almost there! Your photo was a little blurry. Tap for a quick tip to retake it in seconds." Also show an in-app banner with live tips: good lighting, flat surface, no glare.

Day 3 (no retry yet): SMS — higher open rate for lapsed users. "Your account is almost ready. Retake your ID photo in under 60 seconds. [Link]"

Day 5: Activate real-time blur detection in the camera UI so the root cause is fixed for all future users.

Day 7: Final push with social proof: "98% of users who retook their photo were verified in minutes. Your account is waiting!"

Week 2 — Education & Incentive (Days 8–14)

Day 8: Email with a 3-step visual guide (lighting, flat surface, no glare) + a deep-link that opens the camera screen directly.

Day 10: In-app modal on next open: "Welcome back! Your documents are saved. Just retake the photo — you're almost done."

Day 14: Soft urgency push: "Your verification is still pending. Complete it before [Date] to keep your spot."

Week 3 — Final Escalation (Days 15–21)

Day 15: Escalate users with no retry to a support agent for a personalised outreach email.

Day 18: Last SMS: "Final reminder: Your verification expires in 3 days. Retake your photo now. [Link]"

Day 21: Sunset from active campaigns. Flag as 'churned_kyc_failed'. Route to a reactivation campaign after 30 days.

Measuring Success

- **Retry Rate:** % of failed clear error users who attempt KYC again within 21 days. Target: >60%.
- **Retry-to-Pass Rate:** % of retriers who pass. Target: >80%.
- **Channel CTR:** Compare push vs. SMS vs. email to optimise spend.
- **Incremental Activations:** Net new completions directly attributed to the campaign.

A/B test notification copy and timing in Week 1 before scaling the full plan.